

# CardioGuard: Machine Learning System for Heart Disease Prediction Report

## 1. Introduction and Objective

This project aims to develop, evaluate, and deploy a machine learning system, named CardioGuard, for the prediction of heart disease. The primary objective is to build a robust and interpretable model that can assist in early diagnosis, thereby improving patient outcomes. The system leverages clinical features to classify patients into 'No Heart Disease' or 'Heart Disease Present' categories.

## 2. Data Loading and Initial Exploration

The dataset, `heart_disease_uci.csv`, was loaded using `kagglehub`. Initial exploration using `df.head()`, `df.tail()`, and `df.sample()` provided a preliminary understanding of the data structure and content. This step helped in identifying potential issues such as irrelevant columns, incorrect data types, and initial patterns.

## 3. Data Preprocessing and Cleaning

### Data Types and Column Identification

`df.info()` and `df.dtypes` revealed 8 object (categorical) columns and 8 numerical columns. `id` and `dataset` were identified as irrelevant for modeling.

Unique value counts for categorical features were examined to detect inconsistencies or typos.

### Handling Impossible Values

`trestbps` (resting blood pressure) and `chol` (cholesterol) had 0 values, which are medically impossible. These were treated as missing values (`np.nan`) to avoid skewing the data.

### Missing Values Analysis

Significant missingness was observed in `ca` (66.41%), `thal` (52.83%), and `slope` (33.59%), along with other columns like `fbs`, `oldpeak`, `trestbps`, `exang`, `thalch`, `chol`, and `restecg`.

### Imputation Strategy:

Numerical columns (`trestbps`, `chol`, `thalch`, `oldpeak`, `ca`): Imputed with the median value to maintain robustness against outliers.

Categorical/Boolean columns (`fbs`, `restecg`, `exang`, `slope`, `thal`): Imputed with the mode (most frequent value).

### Duplicate Records

No duplicate rows were found in the dataset.

### Target Variable Binarization

The original num target variable, indicating disease severity from 0-4, was binarized into a target column (0 = No Disease, 1 = Disease Present). This transformed the problem into a binary classification task, aligning with clinical utility and common research practices for this dataset.

The binarized target showed a relatively balanced distribution: 44.7% No Disease (0) and 55.3% Disease Present (1).

### Outlier Handling (Winsorization)

Outliers in numerical features (age, trestbps, chol, thalch, oldpeak, ca) were identified using IQR (Interquartile Range) method.

Winsorization (capping at  $1.5 \times \text{IQR}$  bounds) was applied. This approach preserves all patient data while mitigating the distorting effect of extreme values, which are often clinically significant in medical datasets.

## 4. Exploratory Data Analysis (EDA)

### Feature Distributions by Target Class

KDE plots comparing feature distributions for 'No Disease' vs. 'Disease' patients were generated. Features like thalch (lower for disease patients), oldpeak (higher for disease patients), and age (older for disease patients) showed clear separation, indicating strong predictive power.

### Categorical Features vs. Target

Bar plots showed the relationship between each categorical feature and heart disease presence. For instance, 'asymptomatic' chest pain type (cp) showed a high disease rate, highlighting a clinically relevant paradox.

### Correlation Analysis

A correlation heatmap revealed linear relationships between numerical features and with the target variable.

Features with high absolute correlation with the target (exang, cp\_atypical angina, thalch, oldpeak) were noted as strong predictors.

### Pairplot for Top Features

A pairplot of the top 4 most predictive features (oldpeak, thalch, ca, age) visually confirmed the separability of the two classes, suggesting that these features could effectively distinguish between healthy and diseased individuals.

## 5. Feature Engineering

Four new features were engineered based on domain knowledge to capture more complex relationships:

age\_risk\_group: Ordinal variable (0, 1, 2) to capture non-linear age-related risk thresholds.

bp\_chol\_interaction: Product of trestbps and chol to represent combined cardiovascular stress.

exercise\_stress\_score: Combined oldpeak, exang, and normalized thalch to quantify cardiac stress during exercise.

thalch\_age\_ratio: Ratio of achieved maximum heart rate to age-predicted maximum heart rate, indicating cardiac fitness.

These new features showed promising correlations with the target, with exercise\_stress\_score having the highest (0.4777).

## 6. Feature Selection

### Correlation Filter

Features with an absolute correlation to the target less than 0.05 (or undefined correlation due to constant values) were removed. The ca column, despite imputation, showed negligible correlation and was dropped.

### Multicollinearity Check

High multicollinearity (correlation  $> 0.85$ ) was observed between several feature pairs (e.g., thal\_reversible defect and thal\_normal, age\_risk\_group and age). This indicates redundancy. While the current setup passed this check, in a more rigorous approach, one of the highly correlated features might be dropped based on its individual correlation with the target.

The final dataset X retained 21 features after this step.

## 7. Data Preparation for Modeling

### Train-Test Split

The data was split into 80% training and 20% testing sets using train\_test\_split.

Stratification by the target variable ensured that the class distribution (44.7% No Disease, 55.3% Disease) was maintained in both training and testing sets, preventing biased evaluation.

### Feature Scaling

StandardScaler was applied to the numerical features. This transforms data to have a mean of 0 and a standard deviation of 1, which is crucial for distance-based algorithms and models sensitive to feature scales (like Logistic Regression).

Crucially, the scaler was fit\_transform on the training data (X\_train\_scaled) and only transform on the testing data (X\_test\_scaled) to prevent data leakage.

### SMOTE for Class Imbalance

Although the dataset was relatively balanced (44.7% vs 55.3%), SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data only (X\_train\_smote, y\_train\_smote).

SMOTE generated synthetic samples for the minority class ('No Disease') to achieve a perfect 50/50 balance in the training set, which helps prevent model bias towards the majority class.

## 8. Model Training and Evaluation

Three classification models were trained and evaluated:

### Model 1: Logistic Regression (Baseline)

Rationale: A simple, interpretable model to establish a performance baseline. Effective for linear relationships.

Configuration: C=1.0 (moderate regularization), class\_weight='balanced', max\_iter=2000, random\_state=42.

Performance: Training Accuracy: 0.7973, Testing Accuracy: 0.8207, Gap: -0.0234.

### Model 2: Random Forest (Intermediate)

Rationale: An ensemble method that handles non-linear relationships and feature interactions well. More complex than Logistic Regression.

Configuration: n\_estimators=100, max\_depth=10, min\_samples\_leaf=5, class\_weight='balanced', random\_state=42.

Performance: Training Accuracy: 0.8931, Testing Accuracy: 0.7935, Gap: 0.0996.

### Model 3: XGBoost (Advanced)

Rationale: A powerful gradient boosting algorithm widely used for tabular data, known for high performance and regularization capabilities. Preferred over neural networks for this dataset size.

Configuration: `n_estimators=200`, `max_depth=4`, `learning_rate=0.05`, `scale_pos_weight` (calculated based on class imbalance), `subsample=0.8`, `colsample_bytree=0.8`, `random_state=42`.

Performance: Training Accuracy: 0.9496, Testing Accuracy: 0.8315, Gap: 0.1181.

### Regularization Analysis (Logistic Regression)

Tested C values (inverse of regularization strength) from 0.001 to 100.0.

C=1.0 was chosen as it provided the best balance between training and test accuracy, avoiding both underfitting (too strong regularization) and overfitting (too weak regularization).

### Evaluation Metrics (Custom `evaluate_model` function)

All models were evaluated using a comprehensive set of metrics, with special emphasis on medical relevance:

Accuracy: Overall correct predictions.

Precision: Proportion of positive predictions that were actually correct.

Recall (Sensitivity): Proportion of actual positive cases correctly identified. Most critical in medical diagnosis to minimize false negatives.

Specificity: Proportion of actual negative cases correctly identified. Important for avoiding unnecessary further tests for healthy individuals.

F1-Score: Harmonic mean of precision and recall.

AUC (Area Under ROC Curve): Overall model performance across all classification thresholds, robust to class imbalance.

Confusion Matrix: Detailed breakdown of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

### Model Comparison (Initial Test Set Results)

Model Accuracy Precision Recall Specificity F1 AUC

Logistic Regression (Baseline) 0.8207 0.8286 0.8529 0.7805 0.8406 0.8947

Random Forest (Intermediate) 0.7935 0.8077 0.8235 0.7561 0.8155 0.8838

XGBoost (Advanced) 0.8315 0.8381 0.8627 0.7927 0.8502 0.9014

XGBoost demonstrated the best performance across most metrics, particularly AUC.

### ROC Curves

ROC curves showed the trade-off between True Positive Rate and False Positive Rate across various thresholds. XGBoost had the highest AUC (0.901), indicating superior overall discriminative power.

### Bias-Variance Tradeoff Analysis

The analysis compared training vs. test accuracy gaps.

Logistic Regression: Small gap (-0.0234), good fit.

Random Forest: Moderate gap (0.0996), good fit.

XGBoost: Largest gap (0.1181) among the three, but still considered a good fit given the dataset size, indicating a slight bias towards overfitting but good generalization.

## 9. Hyperparameter Tuning (GridSearchCV on XGBoost)

Objective: Optimize XGBoost parameters to maximize roc\_auc.

Method: GridSearchCV with 5-fold cross-validation was used to search across max\_depth ([3, 4, 5]), learning\_rate ([0.01, 0.05, 0.1]), and n\_estimators ([100, 200]).

Best Parameters: {'learning\_rate': 0.1, 'max\_depth': 3, 'n\_estimators': 100}.

Best CV AUC Score: 0.8920.

Tuned XGBoost Test Performance: Test AUC: 0.8981, Test Recall: 0.8431, Test F1: 0.8390.

## 10. Cross-Validation (5-Fold Stratified)

To ensure model robustness and generalizability, 5-fold stratified cross-validation was performed on the training data.

Each fold maintained the class distribution.

Results (Mean AUC):

Logistic Regression: 0.8744 (stable Std Dev: 0.0182)

Random Forest: 0.8804 (stable Std Dev: 0.0141)

XGBoost (Tuned): 0.8930 (stable Std Dev: 0.0158)

Cross-validation confirmed the consistent performance of the models, with Tuned XGBoost showing the highest mean AUC.

## 11. Feature Importance

### XGBoost Feature Importance

The top 5 most important features identified by XGBoost were `exang`, `exercise_stress_score`, `cp_atypical angina`, `thal_normal`, and `sex`.

This aligns well with medical understanding, as these features are known indicators of heart health.

### Logistic Regression Coefficients

Coefficients indicated the direction of influence:

Positive coefficients (e.g., `sex`, `exercise_stress_score`, `age_risk_group`) increased the likelihood of heart disease.

Negative coefficients (e.g., `cp_atypical angina`, `thal_normal`, `cp_non-anginal`) decreased the likelihood.

## 12. Threshold Analysis (XGBoost)

The default classification threshold of 0.5 was analyzed against various thresholds.

For medical applications, Recall (Sensitivity) is paramount to minimize False Negatives (missed disease cases).

A threshold of 0.35 was recommended, which increased Recall from 0.8627 (at 0.5) to 0.9314, while maintaining acceptable Specificity (0.7195).

**Clinical Justification:** A moderate increase in False Positives (leading to further investigation) is preferable to a False Negative (missing a potentially life-threatening condition).

### Precision-Recall Curve

The Precision-Recall curve confirmed the trade-off, showing how precision and recall varied across different thresholds. Tuned XGBoost achieved an Average Precision (AP) of 0.908, indicating good performance for the positive class.

## 13. Unsupervised Learning (K-Means Clustering)

Objective: To explore if distinct patient clusters naturally emerge based on clinical features alone, without using disease labels.

Methodology: Applied K-Means clustering on the scaled training data (without SMOTE or target labels).

Elbow Method: Suggested 2 or 3 as optimal number of clusters, supporting the binary nature of the target.

K-Means with K=2: Identified two main clusters.

Silhouette Score: 0.1327. This low score was interpreted as clinically expected for real-world medical data, as heart disease risk exists on a continuum and patients often fall between clear-cut categories. It does not necessarily indicate poor clustering but rather complex, overlapping patient subgroups.

PCA for Visualization: Reduced data to 2D for plotting, showing some overlap between clusters and actual labels.

Cluster Profile: Each cluster was profiled by the mean values of key features. Cluster 0 showed higher oldpeak, lower thalch, and higher ca, consistent with elevated heart disease risk. Cluster 1 showed an inverse profile, consistent with lower risk. This demonstrated that the algorithm discovered medically meaningful groupings without explicit labels, validating the underlying clinical structure of the features.

#### 14. Model Deployment (Gradio Interface)

A Gradio interface was created to allow interactive prediction using the `xgb_best` model.

The `predict_heart_disease` function encapsulates all preprocessing steps (feature engineering, encoding, scaling) to ensure consistent inference.

The interface provides predicted class, confidence, and probability distribution, using the optimized threshold of 0.35.

#### 15. Conclusion and Key Findings

The CardioGuard ML system effectively predicts heart disease with the Tuned XGBoost model demonstrating superior performance, particularly in Recall and AUC. The optimized threshold of 0.35 was chosen to prioritize minimizing false negatives, which is critical in medical screening.

Key insights from the analysis include:

Effective Preprocessing: Robust handling of missing values and outliers was crucial.

Feature Engineering: Created clinically relevant features that improved model performance.



XGBoost Dominance: Consistently outperformed simpler models on this tabular dataset.

Threshold Optimization: Highlighted the importance of adjusting decision thresholds based on the cost of errors in medical contexts.

Clustering Validation: Unsupervised learning confirmed that the features hold intrinsic clinical meaning, as the algorithm naturally grouped patients into high and low-risk profiles, even with overlapping characteristics typical of real-world medical data.

This project provides a robust and clinically informed approach to heart disease prediction, ready for further validation and potential integration into clinical decision-support systems.